Interventions. Interventional queries concern causal effects from outside a system [70]. An *intervention* is a partial distribution $\mathcal{I}$ over a set of variables $\boldsymbol{V}' \subseteq \boldsymbol{V}$ that replaces each CPD $\Pr(Y \mid \mathbf{Pa}_Y; \theta_Y)$ with a new CPD $\mathcal{I}(Y \mid \mathbf{Pa}^*_Y; \theta^*_Y)$ for each $Y \in \boldsymbol{V}'$. We denote intervened variables by $Y_{\mathcal{I}}$. Interventions are consistent with the causal structure of the graph, i.e., they preserve Markov compatibility.

Example 1 (continued). Let π^S_H be the (honest) type-revealing policy where S retreats if and only if $X = weak$. After the intervention $\mathcal{I}(D^S \mid \mathbf{Pa}_{D^S}; \theta^*_{D^S}) = \pi^S_H$ on D^S which replaces the NE policy for S (to always defend) with π^S_H. T's policy is still a best response (they attack whenever S retreats).

Agents. Kenton et al. [49] define agents as systems that would adapt their policy if their actions influenced the world in a different way. This is the relevant notion of agency for our purposes, as we define belief and intent based on how the agent would adapt its behaviour to such changes. A key assumption is that SCGs are common prior games, the agents in the game share a prior over the variables. We interpret this to mean, additionally, that the agents share the objectively correct prior, that is, their subjective models of the world match reality. This means we are unable to account for cases where an agent intends to deceive someone because they (falsely) believe it is possible to do so.

3 Belief, intention, and deception

We first define belief and extend H&KW's notion of intention. Then we combine these notions to define deception. Our definitions are *functional* [86]: they define belief, intention, and deception in terms of the functional role the concepts play in an agent's behaviour. We provide several examples and results to show that our definitions have desirable properties.

Belief. We take it that agents have beliefs over *propositions*. An *atomic proposition* is an equation of the form $V = v$ for some $V \in \boldsymbol{V}$, $v \in \mathrm{dom}(V)$. A *proposition* is a Boolean formula ϕ of atomic propositions combined with connectives $\neg, \wedge, \vee$. In setting $\boldsymbol{E} = \boldsymbol{e}$ under policy profile $\boldsymbol{\pi}$, an atomic proposition is *true* ($\top$) if the propositional formula is true in that setting, i.e., $X = x$ is true if $X(\boldsymbol{\pi}, \boldsymbol{e}) = x$. The truth-values over Boolean operators are defined in the usual way.

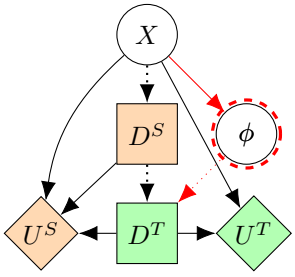


Figure 2: T believes ϕ (Def. 3.1) if 1) they acts as though they observe $\phi = \top$, 2) they would have acted differently if they observed $\phi = \bot$.

We operationalize belief as *acceptance*; essentially, an agent accepts a proposition if they act as though they know it is true [86, 21]. As we argued in section 1, we think that acceptance is the key concept, especially when discussing agents with incentives to influence each other's behaviour. To formalize acceptance of a proposition ϕ, we compare the agent's actual behaviour with its behavior in a game in which the agent can observe ϕ, $\pi^i(\phi) = \pi^i(D^i \mid \mathbf{Pa}_{D^i}, \phi)$. We assume ϕ consists only of variables that are not descendants of D^i so that cycles are not introduced into the graph. For policy profile $\boldsymbol{\pi}$, we assume $\pi^i(\phi)$ is unique given the policies of the other agents: $\boldsymbol{\pi}_{i(\phi)} = (\pi^i(\phi), \boldsymbol{\pi}^{-i})$. The decision the agent would have taken at D^i, had they observed that ϕ were true, can be represented as $D^i_{\phi=\top}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e})$. Importantly, $\phi = \top$ should be understood as only intervening on the agent's observation (and not the proposition itself) as we wish to understand how the agent would have acted, had they believed ϕ, whether or not it was in fact true in the particular setting. In fig. 2 we continue example 1 by allowing T to observe the proposition $\phi : X = strong$ and letting $D^T(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e}) = attack$ if and only if $\phi = \bot$. Clearly ϕ depends on X.

An agent i responds [29] to a proposition ϕ if they act differently when they observe that ϕ is true to when they observe that ϕ is false, i.e., i *responds* to ϕ under $\boldsymbol{\pi}$ in $\boldsymbol{e}$ if $D^i_{\phi=\bot}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e}) \neq D^i_{\phi=\top}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e})$. Then, in words, for a proposition ϕ to which i responds, i believes ϕ if they act as though they observe ϕ is true. If i does not respond to ϕ, then we cannot infer i's belief about ϕ from their behaviour. If they respond to ϕ but do not act as though they observe ϕ is true, then ϕ matters to i, but i does not give ϕ sufficient credence to accept it as a basis for action (they may believe $\neg\phi$ or they may be uncertain).

Definition 3.1 (Belief)**.** Under policy profile $\boldsymbol{\pi} = (\pi^i, \boldsymbol{\pi}^{-i})$, in setting $\boldsymbol{e}$, for agent i and proposition ϕ to which i responds: i *believes* ϕ if i acts as though they observe ϕ is true, i.e., $D^i(\boldsymbol{\pi}, \boldsymbol{e}) = D^i_{\phi=\top}(\boldsymbol{\pi}_{i(\phi)}, \boldsymbol{e})$. An agent has a *true belief* about ϕ if they believe ϕ and ϕ is true (similarly for a false belief). If an agent does not respond to ϕ then its belief about ϕ is unidentifiable from its behaviour.

Example 1 (continued). Under $\boldsymbol{\pi}_{def,\neg att}$, when T observes $\phi : X =$ strong, they attack if and only if S is weak, so T responds to ϕ. Since T never attacks on-equilibrium, they unconditionally act as though $\phi = \top$ (that S is strong). Hence, T always believes ϕ and T has a false belief about ϕ when S is weak.

This definition has nice properties: 1) an agent cannot believe and disbelieve a proposition at once; 2) an agent does not have a false belief about a proposition constituted only by variables they observe. For instance, in example 1, since S observes their type, they never have a false belief about it.

Proposition 3.2 (Belief coherence)**.** *Under policy profile $\boldsymbol{\pi}$ for any agent i, proposition ϕ and setting $\boldsymbol{e}$: 1) i cannot both believe ϕ and $\neg\phi$ in $\boldsymbol{e}$; 2) if i observes every variable constituting ϕ, and i only adapts its policy if observing ϕ changes its subjective probability distribution (i.e. if $P^{\boldsymbol{\pi}}(\cdot \mid pa_{D^i}) = P^{\boldsymbol{\pi}}(\cdot \mid pa_{D^i}, \phi)$ then $\pi^i = \pi^i(\phi)$), then i does not have a false belief about ϕ.*

Intention. Deception is *intentional*. We define the *intention to cause* some outcomes. Our formalisation is related to H&KW's "intent to bring about", but fixes substantive problems with the latter. This account reduces intention to the agent's subjective causal model and utility, given these we can infer an agent's intentions from its behaviour. First, we define a *context-conditional intervention* which only occurs in some settings. We use this to intervene on only the outcomes which the agent intended.

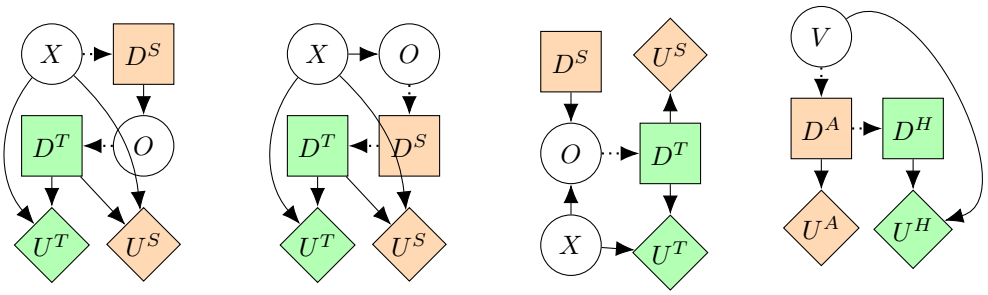
Definition 3.3 (Context-Conditional Intervention)**.** For an intervention $\mathcal{I}$ and set of settings $\boldsymbol{w} \subseteq$ *dom*$(\boldsymbol{E})$, the *context-conditional intervention* on $\boldsymbol{Y}$ is $\boldsymbol{Y}_{\mathcal{I}|\boldsymbol{w}}(\boldsymbol{\pi}, \boldsymbol{e}) = \begin{cases} \boldsymbol{Y}_{\mathcal{I}}(\boldsymbol{\pi}, \boldsymbol{e}) \text{ if } \boldsymbol{e} \in \boldsymbol{w}, \\ \boldsymbol{Y}(\boldsymbol{\pi}, \boldsymbol{e}) \text{ if } \boldsymbol{e} \notin \boldsymbol{w}. \end{cases}$

Informally, an agent intended to cause an outcome $X = x$ with its policy, if guaranteeing that another policy would cause $X = x$ would make that policy just as good for the agent. This follows the spirit of Ashton's counterfactual notion of "desire" as a desiderata for algorithmic intent [4]. For concreteness, in example 1, S intentionally causes $D^T = \neg attack$ with the Nash policy (which always defends) in the settings $\boldsymbol{w}$ in which S is weak. To see this, consider that if T was guaranteed to not attack in the settings in $\boldsymbol{w}$, then the alternate (type-revealing) policy would be just as good for S as the Nash policy. Formally, the context-conditional intervention $D^T_{\boldsymbol{\pi}|\boldsymbol{w}}$ guarantees the desired outcome (no attack) in the settings where S is weak, making the type-revealing policy just as good for S, so S does intend $D^T =$*¬attack* in those settings. We require that $\boldsymbol{w}$ is *subset-minimal*, meaning that no proper subset of $\boldsymbol{w}$ satisfies the condition for intent. For a variable that the agent intentionally influences, the subset-minimality of $\boldsymbol{w}$ excludes unintended outcomes of that variable. For example, when S is strong, they do not intend to cause $D^T = \neg attack$ because in these settings T would not attack regardless of S's policy, so causing these outcomes is not the reason that S chose the Nash policy instead of the type-revealing policy. Below we make this general; following H&KW, we require that $\boldsymbol{X}$ is part of a subset-minimal $\boldsymbol{Y}$ to capture cases in which the agent intends to influence multiple variables. Making the context-conditional intervention in $\boldsymbol{w}$ fixes problems with H&KW's definition (which we discuss in the appendix). Since the agent might intend to cause outcomes of different variables in different settings, we require a different set of settings $\boldsymbol{w}_Y$ for each $Y \in \boldsymbol{Y}$. Additionally, similar to H&KW, we compare the effects of the agent's policy to a set of alternative reference policies to take into consideration the relevant choices available to the agent when it made its decision. In Ward et al. [101], we expand on this formalisation of intention.

Definition 3.4 (Intention)**.** For policy profile $\boldsymbol{\pi} = (\pi^i, \pi^{-i})$, a reference set of alternative policies for i $REF(\pi^i)$, and $\boldsymbol{X} \subseteq \boldsymbol{V}$, agent i *intentionally causes* $\boldsymbol{X}(\boldsymbol{\pi}, \boldsymbol{e})$ with policy π^i if there exists $\hat{\pi}^i \in REF(\pi^i)$, subset-minimal $\boldsymbol{Y} \supseteq \boldsymbol{X}$ and subset-minimal $\boldsymbol{w}_Y \subseteq dom(\boldsymbol{E})$ for each $Y \in \boldsymbol{Y}$ s.t. $\boldsymbol{e} \in \boldsymbol{w}_{\boldsymbol{X}} := \bigcap_{Z \in \boldsymbol{X}} \boldsymbol{w}_Z$ satisfying: $\sum_{U \in \boldsymbol{U}^i} \mathbb{E}_{\boldsymbol{\pi}}[U] \leq \sum_{U \in \boldsymbol{U}^i} \mathbb{E}_{(\hat{\pi}^i, \boldsymbol{\pi}^{-i})}[U_{\{Y_{\boldsymbol{\pi}|\boldsymbol{w}_Y}\}_{Y \in \boldsymbol{Y}}}]$.

Def. 3.4 says that causing the outcomes of the variables in $\boldsymbol{Y}$, in their respective settings $\boldsymbol{w}_Y$, provides sufficient reason to choose π^i over $\hat{\pi}^i$. On the left-hand side (LHS) we have the expected utility to i from playing π^i. The right-hand side (RHS) is the expected utility for agent i under $\hat{\pi}^i$, except that for each $Y \in \boldsymbol{Y}$, in the settings where i intended to cause the outcome of Y, $\boldsymbol{w}_Y$, the outcome of Y is set to the value it would take if i had chosen π^i. The RHS being greater than the LHS means that, if the variables in $\boldsymbol{Y}$ are fixed in their respective settings to the values they would take if π^i were chosen, then $\hat{\pi}^i$ would be at least as good for i. So the *reason* i chooses π^i instead of $\hat{\pi}^i$ is to bring about the values of $\boldsymbol{Y}$ in $\boldsymbol{w}_Y$. We assume that the policies of the other agents are fixed.

Example 2 (Inadvertent misleading fig. 3a). Two submarines must communicate about the location of a mine-field. The signaler S must send the location X to the target T but T only receives a noisy observation O of S's signal. If S honestly signals the location but, due to the noise in the signal, T is



(a) Ex. 2: S *inadvertently* misleads T as T has a noisy observation of D^S.

(b) Ex. 3: An umpire S *mistakenly* misleads T. due to noise.

(c) Ex. 4. S deceives T regarding a proposition about which S is ignorant.

(d) Ex. 5: The agent unintentionally misleads the human as a side-effect.

Figure 3: Inadvertent misleading (3a) and side-effects (3d) are excluded because we require deception to be intentional. Accidental misleading (3b) is not deception because we require that S does not believe ϕ is true.

caused to have a false belief, then S did not deceive T. In this case, causing a false belief was not *intentional*. S intentionally causes T's true beliefs but not T's false beliefs, because the subset-minimal $\boldsymbol{w}_X$ does not contain the settings in which T is caused to have a false belief by the noisy signal.

Def. 3.4 has nice properties: agents do not intentionally cause outcomes they cannot influence.

Proposition 3.5 (Intention coherence)**.** *Suppose $X(\boldsymbol{\pi}_1, \boldsymbol{e}) = X(\boldsymbol{\pi}_2, \boldsymbol{e})$ for all π_1^i and π_2^i with any fixed $\boldsymbol{\pi}^{-i}$. Then i does not intentionally cause $X(\boldsymbol{\pi}, \boldsymbol{e})$ with any policy.*

Theorem 3.6. *If an agent i intentionally causes $X = x$ then D^i is an* actual cause *[41] of $X = x$.*

This follows from the assumption that the agents share the correct causal model (see section 2).

Deception. Following Carson [15], Mahon [58], we say that *an agent S deceives another agent T if S intentionally causes T to believe ϕ, where ϕ is false and S does not believe that ϕ is true.* Formally:

Definition 3.7 (Deception)**.** For agents $S, T \in N$ and policy profile $\boldsymbol{\pi}$, *S deceives T about proposition* ϕ with $\pi^S \in \boldsymbol{\pi}$ in setting $\boldsymbol{e}$ if: 1) S intentionally causes $D^T = D^T(\boldsymbol{\pi}, \boldsymbol{e})$ (with π^S according to def. 3.4); 2) T believes ϕ (def. 3.1) and ϕ is false; 3) S does not believe ϕ.

Condition 1) says that deception is *intentional*. Condition 2) simply says that T is in fact caused to have a false belief. Condition 3) excludes cases in which S is mistaken. In example 1, we showed that S intentionally causes $D^T = \neg attack$, so 1) is satisfied. We already stated 2) that T has a false belief about ϕ when $X = weak$. Finally, as S unconditionally defends, D^S does not respond to ϕ, so S's belief about ϕ is unidentifiable. Therefore, all the conditions for deception are met.

S did not deceive T if S accidentally caused T to have a false belief because S was mistaken.

Example 3 (Mistaken Umpire fig. 3b). A tennis umpire S must call whether a ball X is *out* or *in* to a player T. S's observation O of the ball is 99% accurate. Suppose the umpire believes the ball is *in*, and makes this call, but that they are *mistaken*. This is not deception because condition 3) is not met.

S might deceive about a proposition of which they are ignorant (see supp. material), as motivated by the following example [97] which instantiates Pfeffer and Gal's revealing/denying pattern [73].

Example 4 (Unsafe Bridge fig. 3c). Sarah does not observe the condition of a bridge (X), but she can open a curtain (O) to reveal the bridge to Tim. Tim wants to cross if the bridge is safe but will do so even if he is uncertain. If Sarah knew the bridge was safe, she would cross herself, and if she knew it was unsafe she would reveal this to Tim. Because she is uncertain about the safety of the bridge, she prefers to risk Tim crossing. So, Sarah does not reveal the safety of the bridge which causes Tim to cross. Therefore, when the bridge is unsafe, Sarah has deceived Tim whilst being ignorant.

We adapt Christiano's SmartVault example [19]: an AI system tasked with making predictions about a diamond in a vault unintentionally misleads a human operator as a *side-effect* of pursuing its goal.

Example 5 (Misleading as a side-effect fig. 3d). The variable V determines the location of the diamond (whether it is in the vault or not). The AI agent A observes V but the human H does not. A can either make an incomprehensibly precise prediction of the location of the diamond which cannot be understood by H, or an explainable prediction (just stating the value of V). H has to predict